

2.2. The compact Kähler case. An important class of compact manifolds are those admitting a Kähler metric. This includes all complex submanifolds of projective space.

In this case, the cohomological story greatly simplifies: Roughly speaking, all cohomologies are determined by Dolbeault cohomology. More precisely:

Proposition 2.3 (The $\partial\bar{\partial}$ -Lemma, [DGMS75]). *For any bicomplex $A = (A, \partial, \bar{\partial})$, the following assertions are equivalent:*

- (1) *For any $a \in A$ such that $\partial a = \bar{\partial} a = 0$ and $a = db$ for some $b \in A$, there exists a $c \in A$ such that $a = \partial\bar{\partial}c$.*
- (2) *There is an isomorphism $A \cong A^{sq} \oplus A^{dot}$, where A^{sq} is a direct sum of squares, i.e. bicomplexes of the form*

$$(2.3) \quad \begin{array}{ccc} \mathbb{C} & \longrightarrow & \mathbb{C} \\ \uparrow & & \uparrow \\ \mathbb{C} & \longrightarrow & \mathbb{C}, \end{array}$$

where all arrows are $\pm \text{id}$ and all other maps vanish, and A^{dot} is a direct sum of dots, i.e. one-dimensional bicomplexes with all differentials being zero.

- (3) *All maps in the diagram*

$$(2.4) \quad \begin{array}{ccccc} & & H_{BC}(A) & & \\ & \swarrow & \downarrow & \searrow & \\ H_{\bar{\partial}}(A) & \xRightarrow{\quad} & H_{dR}(A) & \xleftarrow{\quad} & H_{\partial}(A) \\ & \swarrow & \downarrow & \searrow & \\ & & H_A(A) & & \end{array}$$

are isomorphisms.

- (4) *The spectral sequences in the previous diagram degenerate, and the filtrations on de Rham cohomology are n -opposed, i.e. $b_k^{p,q} = 0$ unless $k = p + q$.*

Moreover, for a compact Kähler manifold X , the bicomplex A_X satisfies these conditions.

A complex manifold X for which A_X satisfies the above conditions is called a $\partial\bar{\partial}$ -manifold. Compact Kähler manifolds are $\partial\bar{\partial}$ -manifolds, but the converse is not true. A broader class is for example given by those manifolds bimeromorphic to compact Kähler manifolds, i.e. Fujiki's class $\mathcal{C}$, but also these do not exhaust all $\partial\bar{\partial}$ -manifolds, see e.g. [Fri19], [Li24], [KS23a].

One readily checks that all cohomologies introduced above are compatible with direct sums, vanish on squares and are one-dimensional on dots. Thus, as soon as a cohomology allows to reconstruct the information on the position of the dots, e.g. Dolbeault cohomology, or de Rham cohomology